PHILIPUS SANJAYA

Computer Science Graduate · Artificial Intelligence & Data Science
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SUMMARY

Computer Science graduate with majors in Artificial Intelligence and Data Science (Swinburne Academic Excellence Scholarship recipient), seeking graduate and entry-level software engineering and data/ML roles in Melbourne. Delivered the AWS observability tier of an industry-client capstone, built and audited machine-learning pipelines end-to-end with a focus on reproducibility and honest evaluation, and brings three years of operations and team-leadership experience across Australia and Indonesia.

EDUCATION

Bachelor of Computer Science (Majors: Artificial Intelligence, Data Science) Jul 2023 – Jul 2026
Swinburne University of Technology, Melbourne · Conferred 9 July 2026 · Swinburne Academic Excellence Scholarship

- Relevant units: AI Engineering (MLOps), Applied Machine Learning, AI for Games, Machine Learning, Data Visualisation, Internet of Things Programming, Software Deployment & Evolution, Software Requirements

TECHNICAL PROJECTS

AEGISX — Smart-Grid ICS Anomaly Detection — *Industry capstone, client: Cloak Solutions* 2025 – 2026
AWS (Kinesis, Lambda, S3, SNS, SQS, SageMaker, CloudFormation) · Python · Linear

- Owned the cloud tier in a six-person team as Cloud Computing Specialist: designed and deployed the observability pipeline (Kinesis → Lambda → S3 → SNS, with an SQS dead-letter queue) while live LSTM detection stayed on-premises by design
- Wrote architecture decision records and a CloudFormation template so the environment is reproducible, plus an AWS account-migration runbook for the client
- Managed the team's Linear backlog across 7 sprints and 51 issues; co-presented the cloud approach and project challenges at the final assessment

Sentiment-Driven Stock Predictor — *Coursework rebuilt into a public portfolio project* 2025 – Present
Python · TensorFlow/Keras · FinBERT (Hugging Face) · scikit-learn · pytest · GitHub

- Audited a submitted coursework repo and found plaintext credentials, fabricated README results, data leakage (scaler fitted before the train/test split) and a dataset collapsed to ~23 test samples by API free-tier truncation
- Rebuilt it milestone by milestone: removed ~40% dead code, moved to a src/ layout with one YAML config and one CLI, repaired the test suite, and proved every deterministic artifact bit-identical before and after refactoring
- Evidenced result: adding sentiment features lifted classification F1 from 0.22 to 0.57 on 35 features; leakage fix and migration to the FNSPID dataset in progress

Electricity Demand Forecasting — MLOps Pipeline — *COS40007 group project* 2026
Python · TensorFlow · scikit-learn · DVC · Git

- Built a Random Forest + ANN pipeline forecasting Tetouan Zone 1 electricity demand from weather and time features, with DVC-tracked data and fully deterministic training (metrics reproducible to 11 decimal places)
- Implemented residual-based performance and concept-drift monitoring plus PSI feature-drift detection
- Documented an honest tuning finding: a larger config improved cross-validation R^2 but degraded held-out R^2, so the leaner model was kept

CTF Arena — Multi-Agent Game AI — *COS30002 capstone with coordination research* 2025
Python · pyglet · A · FSM/blackboard · GOAP · steering behaviours*

- Built a 2D multi-agent capture-the-flag simulation with steering, A* pathfinding, finite-state and goal-oriented agents, tracked across 12 milestones in Linear
- Ran a seeded ablation study (~150 matches) showing a coordinating layer won ~98.5% of decisive matches (95% CI 0.95–1.00) through defensive denial; influence-map routing cut agent respawns from 31 to 8 per match
- Caught and corrected a mislabelled confidence interval before submission

Personal Portfolio Site — philsanjaya.com — *Design, build and deployment* 2026
Astro · TypeScript · Tailwind · Cloudflare Pages · GitHub Actions

- Designed and shipped a static portfolio with an animated steering-agent canvas: 12.5 KB gzipped JS against a 50 KB budget, WCAG AA contrast in both themes, and deploys gated on CI checks

- Every published project metric cites its source; unsourced values are shown as pending, never estimated

Facial Recognition Attendance Kiosk — *COS30082 team project* 2025

Python · PyQt · Keras · Git

- Owned system integration, attendance logging with cooldown logic, and a workplace sentiment and attendance dashboard on top of the team's recognition and liveness models

WORK EXPERIENCE

Team Lead — *Nene Chicken, Melbourne* Nov 2025 – Present

- Lead and train team members on shift, and handle day-to-day team issues
- Manage stock and supplier relationships, and escalate operational problems to the owner with proposed solutions

Operations Manager — *Aventru, Indonesia* Jan 2021 – Jul 2023

- Identified and closed loopholes in the stock inventory system; planned inventory levels and oversaw warehouse efficiency
- Trained and supervised new hires on operational procedures

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript, R

Machine learning & data: TensorFlow/Keras, scikit-learn, pandas, NumPy, Hugging Face Transformers, Matplotlib, Seaborn, Plotly

Cloud & DevOps: AWS (Kinesis, Lambda, S3, SNS, SQS, SageMaker, CloudFormation), Docker, GitHub Actions, Cloudflare Pages

IT support: Windows troubleshooting, Microsoft 365 (Outlook, Teams, SharePoint), user accounts, permissions & MFA, device setup, networking fundamentals, technical documentation & runbooks

Engineering practice: Git & Conventional Commits, pytest, DVC, YAML-driven configuration, MLOps (drift monitoring, deterministic training), AI-assisted development (Claude Code), Linear

LEADERSHIP & VOLUNTEERING

Chief Lighting Technician — *Rose of Sharon Church, Melbourne* May 2024 – Present

- Lead a volunteer team for stage lighting at services and events; program cues on a ChamSys QuickQ 10 console; wrote the ministry's safety SOPs and manage its annual equipment budget

Sponsorship Manager — *PPIA Swinburne University* Dec 2024 – Dec 2025

- Secured sponsor funding and support for association events; supported the finance division and coordinated event volunteers

Lighting Technician — *Victory Community Church, Indonesia* Jul 2021 – Jul 2023

- Designed, operated and maintained lighting for live productions on an Avolites Tiger Touch II console; trained new team members